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United States Patent	12405446
Kind Code	B2
Date of Patent	September 02, 2025
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Optical system and image pickup apparatus having the same

Abstract

An optical system images an object with light with a wavelength of 8 μm or higher, and includes a diaphragm and an optical element having an aspherical surface and disposed at a position different from that of the diaphragm. In a section including an optical axis, a thickness of an optical element monotonously increases from an on-axis to an outermost off-axis or the optical element is the thinnest at a position other than an on-axis and an outermost off-axis. A predetermined condition is satisfied.

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Appl. No.:	18/368603
Filed:	September 15, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240012222 A1	Jan. 11, 2024

Foreign Application Priority Data

JP	2020-108208	Jun. 23, 2020
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Related U.S. Application Data

division parent-doc US 17341424 20210608 ABANDONED child-doc US 18368603

Publication Classification

Int. Cl.: G02B9/12 (20060101); G02B5/00 (20060101); G02B9/06 (20060101); H04N23/23 (20230101)

U.S. Cl.:

CPC G02B9/12 (20130101); G02B5/005 (20130101); G02B9/06 (20130101); H04N23/23 (20230101);

Field of Classification Search

CPC: G02B (9/12); G02B (5/005); G02B (9/06); G02B (13/008); G02B (13/14); G02B (13/146); H04N (5/33); H04N (23/20); H04N (23/21); H04N (23/23); H04N (25/20); H04N (25/21)

USPC: 359/355; 359/356; 359/357; 359/350

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Background/Summary

(1) This application is a divisional of U.S. patent application Ser. No. 17/341,424, filed on Jun. 8, 2021, which claims the benefit of and priority to Japanese Patent Application No. 2020-108208, filed on Jun. 23, 2020, each of which is hereby incorporated by reference herein in their entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an optical system compatible with infrared light, for example, one suitable for an image pickup apparatus such as a surveillance camera and an in-vehicle camera.

Description of the Related Art

(2) There is known an optical system (“infrared optical system”) compatible with light in an infrared range (infrared light, infrared ray) (with a wavelength of about 8 μm to 14 μm). Applying the infrared optical system to the image pickup apparatus can visualize thermal information such as a temperature distribution of an object, which is unavailable in the visible wavelength range (with a wavelength of about 0.4 μm to 0.7 μm). Materials (“infrared materials”) that transmit light in the infrared range used for the infrared optical systems include, for example, germanium (Ge), gallium arsenide (GAAS), chalcogenide, zinc selenide (ZnSe), zinc sulfide (ZnS), silicon (Si) and resin (high density polyethylene, etc.) and the like. The infrared optical system is demanded for high optical performance (resolution) in order to detect weak thermal information of a distant object. Japanese Patent Laid-Open No. (“JP”) 10-301024 discloses an infrared optical system having an aspherical surface for correcting various aberrations.

(3) In order to process germanium or silicon into an aspherical shape, difficult processes such as grinding and polishing are required. The optical system disclosed in JP 10-301024 uses a flat lens made of thin flat silicon in order to reduce aspherical processing difficulty, but is silent about a shape of the flat lens so as to obtain the high optical performance. The infrared materials other than germanium and silicon can provide an aspherical surface by molding, which is less difficult than grinding or polishing. However, this moldable infrared material has a large dispersion, and it is thus necessary to properly set the focal lengths of the overall system and each lens in order to obtain the high optical performance.

SUMMARY OF THE INVENTION

(4) The present invention provides an optical system that is easy to manufacture and has high optical performance in an infrared range, and an image pickup apparatus having the same.

(5) An optical system according to one aspect of the present invention images an object with light with a wavelength of 8 μm or longer, and includes an optical element having an aspherical surface and disposed at a position different from that of a diaphragm. In a section including an optical axis, a thickness of an optical element monotonously increases from an on-axis to an outermost off-axis or the optical element is the thinnest at a position other than an on-axis and an outermost off-axis (between an on-axis and an outermost off-axis not inclusive). The following conditional expression is satisfied:

$$0.0 < |f/Pf1| < 0.3$$

where f is a focal length of the optical system, and $Pf1$ is a focal length of the optical element.

Alternatively, the following conditional expressions are satisfied:

$$20 \leq (N10 - 1)/(N8 - N12) \leq 800$$

$$0.0 < |f/Pf2| < 0.3$$

where $N8$ is a refractive index of a material of an optical element at a wavelength of 8 μm , $N10$ is a refractive index of the material at a wavelength of 10 μm , $N12$ is a refractive index of the material at a wavelength of 12 μm , f is a focal length of the optical system, and $Pf2$ is a focal length of the optical element.

(6) An image pickup apparatus that includes the above optical system also constitutes another aspect of the present invention.

(7) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a sectional view of an optical system according to Example 1.
- (2) FIG. 2 is an MTF diagram of the optical system according to Example 1.
- (3) FIG. 3 is a sectional view of an optical system according to Example 2.
- (4) FIG. 4 is an MTF diagram of the optical system according to Example 2.
- (5) FIG. 5 is a sectional view of an optical system according to Example 3.
- (6) FIG. 6 is an MTF diagram of the optical system according to Example 3.
- (7) FIG. 7 is a sectional view of an optical system according to Example 4.
- (8) FIG. 8 is an MTF diagram of the optical system according to Example 4.
- (9) FIG. 9 is a sectional view of an optical system according to Example 5.
- (10) FIG. 10 is an MTF diagram of the optical system according to Example 5.
- (11) FIG. 11 is a sectional view of an optical system according to Example 6.
- (12) FIG. 12 is an MTF diagram of the optical system according to Example 6.
- (13) FIG. 13 is a sectional view of an optical system according to Example 7.
- (14) FIG. 14 is an MTF diagram of the optical system according to Example 7.
- (15) FIG. 15 is a sectional view of an optical system according to Example 8.
- (16) FIG. 16 is an MTF diagram of the optical system according to Example 8.
- (17) FIG. 17 is a sectional view of an optical system according to Example 9.
- (18) FIG. 18 is an MTF diagram of the optical system according to Example 9.
- (19) FIG. 19 is a sectional view of an optical system according to Example 10.
- (20) FIG. 20 is an MTF diagram of the optical system according to Example 10.
- (21) FIG. 21 is a sectional view of an optical system according to Example 11.
- (22) FIG. 22 is an MTF diagram of the optical system according to Example 11.
- (23) FIG. 23 is a sectional view of an optical system according to Example 12.
- (24) FIG. 24 is an MTF diagram of the optical system according to Example 12.
- (25) FIG. 25 is a schematic view of principal part of a camcorder as an illustrative image pickup apparatus according to Example 13.

DESCRIPTION OF THE EMBODIMENTS

(26) Referring now to the accompanying drawings, a detailed description will be given of embodiments according to the present invention. Corresponding elements in respective figures will be designated by the same reference numerals, and a duplicate description thereof will be omitted.

(27) A silicon material or a germanium material in the description of each example means a material containing silicon or germanium as a main component, and is not limited to a material exclusively consisting of silicon (Si) or germanium (Ge) but may contain a small amount of impurities.

Example 1

(28) FIG. 1 is a sectional view of an optical system **100** according to this example. The optical system **100** is an infrared optical system having a focal length of 18 mm and an F-number of 0.8. The infrared optical system, as used herein, is an optical system that images an object with light having a wavelength of 8 μm or longer. The optical system **100** includes, in order from the object side to the image side, a first lens **L11** having a positive refractive power and made of a silicon material, a thin aspherical plate (optical element) **P11** made of a silicon material and having an aspherical portion with a thickness that gradually increases from the center to the periphery (or with a thickness that monotonously increases from the on-axis to the outermost off-axis in the

section including the optical axis), a diaphragm (aperture stop) **S1**, a second lens **L12** having a positive refractive power and made of a silicon material, a third lens **L13** having a positive refractive power and made of a silicon material, and a thin aspherical plate (optical element) **P12** made of a silicon material and having an aspherical portion with a thickness that gradually increases from the center to the periphery. The thin aspherical plates **P11** and **P12** are assumed to have a very thin thickness of several hundred μm . Therefore, it can be manufactured by easy processing such as processing used to manufacture a Schmidt correction plate. Light in the infrared range (with a wavelength 8 to 14 μm) guided by the optical system **100** passes through a cover glass **CG1** and forms an image on an infrared sensor **IM1**. A window material may be provided between the first lens **L11** and the object, or the cover glass **CG1** may include an infrared material other than the germanium material. Table 1 shows numerical data of the optical system **100**. The unit of radius of curvature and spacing (distance or interval) is mm

(29) TABLE-US-00001 TABLE 1 RADIUS OF GLASS CURVATURE SPACING MATERIAL
 OBJECT PLANE — INFINITY L11 SPHERICAL SURFACE 16.30 1.50 SILICON SPHERICAL
 SURFACE 17.91 5.76 FLAT SURFACE FLAT SURFACE 0.20 SILICON ASPHERICAL
 SURFACE 11 1.00E+16 4.16 S2 FLAT SURFACE — 4.10 L12 SPHERICAL SURFACE 10.19
 1.60 SILICON SPHERICAL SURFACE 9.44 3.52 L13 SPHERICAL SURFACE 17.60 1.40
 SILICON SPHERICAL SURFACE 27.55 1.82 P2 ASPHERICAL SURFACE 12 1.00E+16 0.20
 SILICON FLAT SURFACE FLAT SURFACE 2.00 CG1 FLAT SURFACE FLAT SURFACE 1.00
 GERMANIUM FLAT SURFACE FLAT SURFACE 3.60 IMAGE PLANE

(30) Table 2 shows aspherical shape data.

(31) TABLE-US-00002 TABLE 2 ASPHERICAL ASPHERICAL SURFACE 11 SURFACE 12
 PARAXIAL RADIUS OF 1.00E+16 1.00E+16 CURVATURE R CONICAL COEFFICIENT k
 0.00 0.00 4TH-ORDER COEFFICIENT A 1.15E-05 -3.67E-05 6TH-ORDER
 COEFFICIENT B -7.80E-08 1.24E-07 8TH-ORDER COEFFICIENT C 1.20E-09 7.32E-09
 10TH-ORDER COEFFICIENT D -1.07E-11 -6.89E-10 12TH-ORDER COEFFICIENT E
 4.02E-14 1.36E-11 14TH-ORDER COEFFICIENT F -3.76E-17 -8.20E-14

(32) The aspherical shape is expressed as follows:

$$(33) \quad Z = \frac{(1/R)h^2}{1 + \sqrt{1 - (1+k)(1/R)^2 h^2}} + Ah^4 + Bh^6 + Ch^8 + Dh^{10} + Eh^{12} + Fh^{14} \quad (1)$$

where Z is a displacement amount from a surface apex in an optical axis direction, h is a height from an optical axis in a direction orthogonal to the optical axis, R is a paraxial radius of curvature, k is a conical coefficient, and A to F are fourth to fourteenth order aspherical coefficients.

(34) FIG. 2 is an MTF diagram of the optical system **100**. A pixel pitch of a general infrared sensor is several tens of μm . This example uses the infrared sensor **IM1** having a pixel pitch of 17 μm by an implementation. In this case, the Nyquist frequency is about 30 lp/mm. In order to resolve the object at this Nyquist frequency, the MTF value of about 30% is empirically sufficient. The MTF value at the Nyquist frequency in this example represented by reference numeral **11** in FIG. 2 is 28%. Since the MTF value is about 30% at the Nyquist frequency of the infrared sensor **IM1**, the optical system **100** exhibits excellent optical performance.

(35) The silicon material has a high refractive index and a low dispersion in the infrared range, and can provide high optical performance with an aspherical surface as part of the lens made of the silicon material. In order to process an optical element having an aspherical surface of a silicon material, a highly difficult process such as grinding or polishing is required. Accordingly, a thin aspherical plate is useful in which it is easy to process an aspherical portion without any highly difficult processes such as grinding or polishing. The thin aspherical plate can be manufactured by a photolithography process and a process for the Schmidt correction plate, which is manufactured by adsorbing an aspherical prototype on a thin lens substrate, by transferring the shape of the aspherical prototype, and by polishing it.

(36) Table 3 shows a refractive index N10 and an Abbe number ν_{10} of a material that transmits

infrared light. The refractive index N_{10} is a refractive index at a wavelength of $10\text{ }\mu\text{m}$. The Abbe number v_{10} is expressed by the following expression (2) where N_8 is a refractive index of the material at a wavelength of $8\text{ }\mu\text{m}$ and N_{12} is a refractive index of the material at a wavelength of $12\text{ }\mu\text{m}$. In general, the larger the Abbe number is, the smaller the refractive index change (dispersion) becomes due to the wavelength. Since numerical values are slightly different depending on each glass material manufacturer, Table 3 shows approximate numerical values. In particular, the refractive index and Abbe number of chalcogenides vary greatly depending on the type of compound using the oxygen group element of Group 16 in the periodic table. Depending on the type of compound, the refractive index N_{10} varies from 2.0 to 4.0 and the Abbe number v_{10} varies from 100 to 800.

(37) TABLE-US-00003 TABLE 3 ZINC ZINC GERMANIUM SILICON RESIN
CALCOGENIDE SULFIDE SELENIDE REFRACTIVE 4.0 3.4 1.5 2.5 2.2 2.4 INDEX N_{10}
DISPERSION 861 1860 45 109 23 57 VALUE v_{10}

$$(38) \quad v_{10} = \frac{N_{10} - 1}{N_8 - N_{12}} \quad (2)$$

(39) Germanium and silicon each have a higher refractive index and a smaller dispersion than other materials. A general N-unit optical system has a configuration that satisfies the following expression (3) in order to correct the chromatic aberration. In the expression (3), f_1 , f_2 , f_3 , $\dots$, and f_n are focal lengths of a first lens, a second lens, a third lens, $\dots$, and an n-th lens, respectively, and v_1 , v_2 , v_3 , $\dots$, and v_n are Abbe numbers of the first lens, the second lens, the third lens, $\dots$, and the n-th lens, respectively.

$$(40) \quad \frac{1}{f_1 v_1} + \frac{1}{f_2 v_2} + \frac{1}{f_3 v_3} + \dots + \frac{1}{f_n v_n} = 0 \quad (3)$$

(41) A lens usually has a positive Abbe number, and thus at least one lens may have a negative focal length in order to reduce the chromatic aberration. Therefore, the optical system for correcting the chromatic aberration has a combination of a positive lens and a negative lens. Since the silicon lens has a very small dispersion, the chromatic aberration is small even with the silicon lens having a positive refractive power alone, but there may be an element that plays a role of a negative lens. In this example, the thin aspherical plate has an aspherical portion with a thickness that gradually increases from the center to the periphery, so that the chromatic aberration can be corrected with high accuracy. An optical system having a short focal length needs to converge light beams with a plurality of angles of view, so that the first lens may be a negative lens.

(42) In order to correct the curvature of field, the optical system needs to satisfy the following expression (4) so as to reduce the Petzval sum. Since the Petzval sum correlates with the curvature of field, the curvature of field can be reduced by reducing the Petzval sum. In the expression (4), f_1 , f_2 , f_3 , $\dots$, and f_n are focal lengths of the first lens, the second lens, the third lens, $\dots$, and the n-th lens, respectively, and N_1 , N_2 , N_3 , $\dots$, and N_n are refractive indexes of the first lens, the second lens, the third lens, $\dots$, and the n-th lens, respectively.

$$(43) \quad \frac{1}{f_1 N_1} + \frac{1}{f_2 N_2} + \frac{1}{f_3 N_3} + \dots + \frac{1}{f_n N_n} = 0 \quad (4)$$

(44) A lens usually has a positive refractive index, and thus at least one lens may have a negative focal length in order to reduce the Petzval sum. Therefore, the optical system for correcting a curvature of field has a combination of a positive lens and a negative lens. Since a silicon lens has a very large refractive index, the Petzval sum can be reduced only by the silicon lens having a positive refractive power, but there may be an element that plays a role of a negative lens. In this example, the Petzval sum can be corrected with high accuracy by the thin aspherical plate having an aspherical portion with a thickness that gradually increases from the center to the periphery. An optical system having a short focal length needs to converge light beams with a plurality of angles of view, so that the first lens may be a negative lens.

(45) For high optical performance, it is important to correct a spherical aberration with high accuracy that is proportional to the pupil diameter of the lens. It is useful to share the spherical aberration with each lens and to correct it. In each example, the spherical lens having the main

refractive power may be a positive lens. Thereby, the light beams can be gently converged, and the spherical aberration can be suppressed. An optical system having a short focal length needs to converge light beams with a plurality of angles of view, so that the first lens may be a negative lens. (46) A bright F-number lens such as an infrared lens needs to correct a high-order curvature of field with high accuracy in order to obtain excellent imaging performance. Accordingly, the present invention places the thin aspherical plate P11 at a position different from that of the diaphragm S1, and causes the aspherical portion of the thin aspherical plate P11 to correct the higher-order curvature of field. Each example places the thin aspherical plate P12 having a small refractive power at a position close to the diaphragm S1 on the object side or the image side of the diaphragm S1 and corrects a higher-order curvature of field and spherical aberration. That is, the conditions for Petzval sum and chromatic aberration correction are shared with spherical lenses having main refractive powers, and a thin aspherical plate having a small refractive power corrects a higher-order spherical aberration and a higher-order curvature of field. It is particularly difficult to correct the high-order curvature of field only with a spherical lens, and thus a thin aspherical plate may be disposed at a position where the on-axis ray and the off-axis ray are separated. Due to this configuration, the optical system 100 can exhibit high optical performance.

(47) Compared to other spherical lenses, the focal length of the thin aspherical plate gives only the action of the aspherical portion as an effect, so it is unnecessary to have a high refractive power. Therefore, the optical systems according to Examples 1 to 8 satisfy the following conditional expression (5):

$$0.0 < |f/Pf1| < 0.3 \quad (5)$$

where f is a focal length of the optical system and $Pf1$ is a focal length of the thin aspherical plate.

(48) The numerical range of the conditional expression (5) may be set to that of the following conditional expression (5a):

$$0.00 < |f/Pf1| < 0.25 \quad (5a)$$

(49) As described above, the configuration according to this example can realize an optical system that is easy to manufacture and has high optical performance in the infrared range.

(50) In the optical systems according to Examples 1 to 8, the thin aspherical plate may have a thin shape from the viewpoint of the transmittance of the silicon material. Hence, the following conditional expression (6) may be satisfied:

$$0.05 < T < 1.00 \quad (6)$$

where T [mm] is a central thickness (thickness on the optical axis).

(51) If the center thickness T is located out of the range of the conditional expression (6), the transmittance may be significantly reduced and the optical performance may be deteriorated.

(52) The numerical range of the conditional expression (6) may be set to that of the following conditional expression (6a):

$$0.05 \leq T \leq 0.60 \quad (6a)$$

(53) The optical systems according to Examples 1 to 4 may satisfy at least of the following conditional expressions (7) to (9):

$$0.1 < |f1/f| < 6.0 \quad (7)$$

$$0.1 < f2/f < 20.0 \quad (8)$$

$$0.1 < f3/f < 5.0 \quad (9)$$

where $f1$, $f2$, and $f3$ are focal lengths of the first to third lenses L11, L12, and L13, respectively.

(54) The conditional expressions (7) to (9) are set to the optical systems according to Examples 1 to 4 so as to satisfactorily correct various aberrations such as the chromatic aberration, the spherical aberration, and the curvature of field and to exhibit high optical performance. If the value is higher than the upper limit or lower than the lower limit in each of the conditional expressions (7) to (9), a correction balance between the curvature of field and the spherical aberration destroys, and the optical performance deteriorates.

(55) The numerical ranges of the conditional expressions (7) to (9) may be set to those of the

following conditional expressions (7a) to (9a):

$$0.1 < |f1/f| < 5.0 \quad (7a)$$

$$0.5 < f2/f < 15.0 \quad (8a)$$

$$0.1 < f3/f < 4.0 \quad (9a)$$

(56) Table 4 shows numerical values corresponding to the conditional expressions according to Examples 1 to 4.

(57) TABLE-US-00004

TABLE	4	f	f1	f2	f3	Pf1	Pf2	f1/f	f2/f	f3/f	f/Pf1	f/Pf2	EX.
1	16	45.3	104.7	18.3	0	0	2.52	5.82	1.02	—	—	—	EX. 2
2	14	57.5	147.6	13.2	0	0	4.11	10.54	0.95	—	—	—	EX. 3
3	50	70.6	59.7	—	0	0	1.41	1.19	—	—	—	—	EX. 4
4	6	—	14.16	11.3	15.08	51.7	49.6	—	2.36	1.88	2.51	0.011805	
												0.012097	

Example 2

(58) FIG. 3 is a sectional view of an optical system **200** according to this example. The optical system **200** is an infrared optical system having a focal length of 14 mm and an F-number of 0.8. The optical system **200** includes, in order from the object side to the image side, a first lens **L21** having a positive refractive power and made of a silicon material, a thin aspherical plate **P21** made of a silicon material and having an aspherical portion with a thickness that gradually increases from the center to the periphery, a thin aspherical plate **P22** made of a silicon material and having an aspherical portion with a thickness that gradually decreases from the center to the periphery, a diaphragm **S2**, a second lens **L22** having a positive refractive power and made of a silicon material, a thin lens **L23** having a positive refractive power and made of a silicon material, and a thin aspherical plate **P23** made of a silicon material and having an aspherical portion with a thickness that gradually increases from the center to the periphery. The thin aspherical plates **P21**, **P22**, and **P23** are assumed to have a very thin thickness of several hundred μm . Therefore, each of them can be manufactured by easy processing such as processing used to manufacture the Schmidt correction plate. This example disposes a thin aspherical plate **P22** in order to correct a high-order spherical aberration. The light in the infrared range guided by the optical system **200** passes through a cover glass **CG2** and forms an image on an infrared sensor **IM2**. Table 5 shows numerical data of the optical system **200**. The unit of radius of curvature and spacing is mm.

(59) TABLE-US-00005

TABLE	5	RADIUS OF GLASS CURVATURE	SPACING	MATERIAL
OBJECT	—	INFINITY	PLANE	L21 SPHERICAL SURFACE 16.30 1.20 SILICON SPHERICAL SURFACE 17.91 3.74
P21	FLAT SURFACE	FLAT SURFACE	0.20	SILICON ASPHERICAL SURFACE 21 1.00E+16 3.42
P22	FLAT SURFACE	FLAT SURFACE	0.70	SILICON ASPHERICAL SURFACE 22 1.00E+16 0.83
S2	FLAT SURFACE	—	1.43	L22 SPHERICAL SURFACE 10.19 1.60
SILICON SPHERICAL SURFACE	9.44 2.87	L23 SPHERICAL SURFACE 17.60 1.40	SILICON SPHERICAL SURFACE 27.55 2.23	P23 ASPHERICAL SURFACE 23 1.00E+16 0.20
SILICON FLAT SURFACE	FLAT SURFACE 2.00	CG2 FLAT SURFACE	FLAT SURFACE 1.00	GERMANIUM FLAT SURFACE
FLAT SURFACE	FLAT SURFACE 3.00	IM2		

(60) Table 6 shows aspherical shape data. The aspherical shape is represented by the expression (1).

(61) TABLE-US-00006

TABLE	6	ASPHERICAL	ASPHERICAL	ASPHERICAL	SURFACE 21
SURFACE 22	SURFACE 23	PARAXIAL RADIUS	1.00E+18	1.00E+18	1.00E+18
OF CURVATURE R	CONICAL COEFFICIENT k	0.00	0.00	0	4TH-ORDER COEFFICIENT A
4.19E-05	4.65E-05	-0.00012	6TH-ORDER COEFFICIENT B	2.508-08	1.72E-07
-1.04E-06	8TH-ORDER COEFFICIENT C	1.63E-10	5.57E-09	1.02E-07	10TH-ORDER COEFFICIENT D
6.34E-11	7.43E-11	-3.42E-09	12TH-ORDER COEFFICIENT E	-9.48E-13	6.76E-13
5.33E-11	14TH-ORDER COEFFICIENT F	6.30E-15	2.63E-15	-3.01E-13	

(62) FIG. 4 is an MTF diagram of the optical system **200**. This example uses the infrared sensor **IM2** having a pixel pitch of 17 μm as an implementation. In this case, the Nyquist frequency is about 30 lp/mm. The MTF value at the Nyquist frequency in this example represented by reference numeral **21** in FIG. 4 is 43%. Since the MTF value is 30% or higher at the Nyquist frequency of the

infrared sensor IM2, the optical system **200** exhibits excellent optical performance.

Example 3

(63) FIG. 5 is a sectional view of an optical system **300** according to this example. The optical system **300** is an infrared optical system having a focal length of 50 mm and an F-number of 0.8. The optical system **300** includes, in order from the object side to the image side, a first lens L31 made of a silicon material and having a positive refractive power, a thin aspherical plate P31 made of a silicon material and having an aspherical portion with a thickness that gradually increases from the center to the periphery, a diaphragm S3, a second lens L32 having a positive refractive power and made of a silicon material, and a thin aspherical plate P32 made of a silicon material and having an aspherical portion with a thickness that gradually increases from the center to the periphery. Therefore, it can be manufactured by easy processing such as a bending process used to manufacture the Schmidt correction plate. The thin aspherical plates P31 and P32 are assumed to have a very thin thickness of several hundred μm . Light in the infrared range guided by the optical system **300** passes through a cover glass CG3 and forms an image on an infrared sensor IM3. Table 7 shows numerical data of the optical system **300**. The unit of radius of curvature and spacing is mm.

(64) TABLE-US-00007 TABLE 7 RADIUS OF GLASS CURVATURE SPACING MATERIAL
OBJECT — INFINITY PLANE L31 SPHERICAL SURFACE 16.30 3.5 SILICON
SPHERICAL SURFACE 17.91 9.0 P31 FLAT SURFACE FLAT SURFACE 0.3 SILICON
ASPHERICAL SURFACE 31 1.00E+16 0.7 S3 FLAT SURFACE FLAT SURFACE 31.6 L32
SPHERICAL SURFACE 52.95227 1.8 SILICON SPHERICAL SURFACE 82.412 14.0 P32
ASPHERICAL SURFACE 32 1.00E+16 0.2 SILICON FLAT SURFACE FLAT SURFACE 3.7
CG3 FLAT SURFACE FLAT SURFACE 1.0 GERMANIUM FLAT SURFACE FLAT SURFACE
3.0 IM3

(65) Table 8 shows aspherical shape data. The aspherical shape is represented by the expression (1).

(66) TABLE-US-00008 TABLE 8 ASPHERICAL ASPHERICAL SURFACE 31 SURFACE 32
PARAXIAL RADIUS OF 1.00E+18 1.00E+18 CURVATURE R CONICAL COEFFICIENT k
0.00 0.00 5TH-ORDER COEFFICIENT A 3.27E-07 -7.33E-06 6TH-ORDER
COEFFICIENT B -1.17E-10 3.85E-08 8TH-ORDER COEFFICIENT C 4.20E-14 9.48E-11
10TH-ORDER COEFFICIENT D 2.11E-17 -1.46E-11 12TH-ORDER COEFFICIENT E
6.93E-21 1.69E-13 14TH-ORDER COEFFICIENT F 0.00E+00 -5.85E-16

(67) FIG. 6 is an MTF diagram of the optical system **300**. This example uses the infrared sensor IM3 having a pixel pitch of 17 μm as an implementation. In this case, the Nyquist frequency is about 30 lp/mm. The MTF value at the Nyquist frequency of this example represented by reference numeral **31** in FIG. 6 is 32%. Since the MTF value is 30% or higher at the Nyquist frequency of the infrared sensor IM3, the optical system **300** exhibits excellent optical performance.

Example 4

(68) FIG. 7 is a sectional view of an optical system **400** according to this example. The optical system **400** is an infrared optical system having a focal length of 6 mm and an F-number of 0.9. The optical system **400** includes, in order from the object side to the image side, a first lens L41 having a negative refractive power and made of a silicon material, a thin aspherical plate P41 made of a silicon material and having an aspherical portion with a thickness that gradually increases from the center to the periphery, a second lens L42 having a positive refractive power and made of a silicon material, a diaphragm S4, a third lens L43 having a positive refractive power and made of a silicon material, and a thin aspherical plate P42 made of a silicon material and having an aspherical portion with a thickness that gradually increases from the center to the periphery. Therefore, it can be manufactured by easy processing such as a bending process used to manufacture the Schmidt correction plate. The thin aspherical plates P41 and P42 are assumed to have a very thin thickness of several hundred μm . Light in the infrared range guided by the optical system **400** passes through a cover glass CG4 and forms an image on an infrared sensor IM4. Table 9 shows numerical data of

the optical system **400**. The unit of radius of curvature and spacing is mm.

(69) TABLE-US-00009 TABLE 9 RADIUS OF GLASS CURVATURE SPACING MATERIAL
OBJECT INFINITY PLANE L41 SPHERICAL SURFACE 9.745272 1.0 SILICON
SPHERICAL SURFACE 7.033832 6.3 P41 ASPHERICAL SURFACE 41 1249.747 0.2
SILICON FLAT SURFACE FLAT SURFACE 5.4 L42 SPHERICAL SURFACE -155.564 1.5
SILICON SPHERICAL SURFACE -2.34E+01 0.6 S4 FLAT SURFACE FLAT SURFACE 6.5 L43
SPHERICAL SURFACE 51.53527 1.0 SILICON FLAT SURFACE -122.591 1.3 P42
ASPHERICAL SURFACE 42 1200.627 0.2 SILICON FLAT SURFACE FLAT SURFACE 2.0
CG4 FLAT SURFACE FLAT SURFACE 1.0 GERMANIUM FLAT SURFACE FLAT SURFACE
3.0 IM4

(70) Table 10 shows aspherical shape data. The aspherical shape is represented by the expression (1).

(71) TABLE-US-00010 TABLE 10 ASPHERICAL ASPHERICAL SURFACE 41 SURFACE 42
PARAXIAL RADIUS OF 1.25E+03 1.20E+03 CURVATURE R CONICAL COEFFICIENT k
0.00 0.00 4TH-ORDER COEFFICIENT A -7.89E-05 -7.80E-05 6TH-ORDER
COEFFICIENT B -2.17E-07 -1.43E-06 8TH-ORDER COEFFICIENT C -1.46E-08
1.44E-07 10TH-ORDER COEFFICIENT D 5.68E-10 -6.54E-09 12TH-ORDER
COEFFICIENT E -8.71E-12 1.36E-10 14TH-ORDER COEFFICIENT F 8.49E-14 -1.10E-12

(72) FIG. 8 is an MTF diagram of the optical system **400**. This example uses the infrared sensor IM4 having a pixel pitch of 17 pin as an implementation. In this case, the Nyquist frequency is about 30 lp/mm. The MTF value at the Nyquist frequency in this example represented by reference numeral **41** in FIG. 8 is 36%. Since the MTF value is 30% or higher at the Nyquist frequency of the infrared sensor IM4, the optical system **400** exhibits excellent optical performance.

Example 5

(73) FIG. 9 is a sectional view of an optical system **500** according to this example. The optical system **500** is an infrared optical system having a focal length of 4.5 mm and an F-number of 0.8. The optical system **500** includes, in order from the object side to the image side, a first lens L51 having a negative refractive power and made of a silicon material, a thin aspherical plate P51 made of a silicon material and having an aspherical portion that is the thinnest between the center and the outermost periphery not inclusive (with a thickness that becomes the thinnest at a position other than an on-axis and an outermost off-axis (between the on-axis and the outermost off-axis not inclusive) in a section including the optical axis), a second lens L52 having a positive refractive power and made of a silicon material, a diaphragm S5, and a third lens L53 having a positive refractive power and made of a silicon material. The thin aspherical plates P51 and P52 are assumed to have a very thin thickness of several hundred μm . Therefore, it can be manufactured by easy processing such as a bending process used to manufacture the Schmidt correction plate. Light in the infrared range guided by the optical system **500** passes through a cover glass CG5 and forms an image on an infrared sensor IM5. Table 11 shows numerical data of the optical system **500**. The unit of radius of curvature and spacing is mm.

(74) TABLE-US-00011 TABLE 11 RADIUS OF GLASS CURVATURE SPACING MATERIAL
OBJECT — INFINITY PLANE L51 SPHERICAL SURFACE 8.63 0.45 SILICON
SPHERICAL SURFACE 6.31 10.48 P51 ASPHERICAL SURFACE ST 196.43 0.20 SILICON
FLAT SURFACE FLAT SURFACE 7.22 L52 SPHERICAL SURFACE 4.39E-01 1.50 SILICON
SPHERICAL SURFACE -22.31 0.10 S5 FLATSURFACE FLATSURFACE 4.04 L53
SPHERICAL SURFACE 14.84 1.65 SILICON SPHERICAL SURFACE 24.16 2.26
SPHERICAL SURFACE FLAT SURFACE 2.00 FLAT SURFACE FLAT SURFACE 1.00
GERMANIUM CG5 FLAT SURFACE FLAT SURFACE 2.00 FLATSURFACE FLAT SURFACE
2.10 IM5

(75) Table 12 shows aspherical shape data. The aspherical shape is represented by the expression (1).

(76) TABLE-US-00012 TABLE 12 ASPHERICAL SURFACE 51 PARAXIAL RADIUS OF CURVATURE R 1.96E+02 CONICAL COEFFICIENT k 0.00 4TH-ORDER COEFFICIENT A -4.11E-05 6TH-ORDER COEFFICIENT B 2.12E-07 8TH-ORDER COEFFICIENT C 0.00E+00 10TH-ORDER COEFFICIENT D 0.00E+00 12TH-ORDER COEFFICIENT E 0.00E+00 14TH-ORDER COEFFICIENT F 0.00E+00

(77) FIG. 10 is an MTF diagram of the optical system 500. This example uses the infrared sensor IM5 having a pixel pitch of 17 pin as an implementation. In this case, the Nyquist frequency is about 30 lp/mm. The MTF value at the Nyquist frequency in this example represented by reference numeral 51 in FIG. 10 is 48%. Since the MTF value is 30% or higher at the Nyquist frequency of the infrared sensor IM5, the optical system 500 exhibits excellent optical performance.

(78) One method of correcting a curvature of field with high accuracy is a method of placing a thin aspherical plate having an aspherical effect at a position where light rays at respective angles of view are separated. The position where the light rays at respective angles of view are separated is different from the position of the diaphragm. Scattering of rays at respective angles of view is corrected by the aspherical surface portion. Since the curvature of field correction up to the fourth-order aspherical term is not enough, it is necessary to have an aspherical shape represented by a function having a sixth-order or higher aspherical term. It is necessary for the aberration correction that the shape has the thinnest thickness between the center and the outermost periphery (not inclusive).

(79) The thin aspherical plate may be disposed at a position different from the position where the diaphragm is disposed from the viewpoint of curvature of field correction. The following expression (10) may be satisfied when the thin aspherical plate is disposed on the object side of the diaphragm:

$$PZ/LP < 0.8 \quad (10)$$

where LP is a distance from the first lens to the diaphragm, and PZ is a distance from the diaphragm to the thin aspherical plate.

(80) When the thin aspherical plate is disposed on the image side of the diaphragm, the following conditional expression (11) may be satisfied:

$$PZ/LS < 0.8 \quad (11)$$

where LS is a distance from the diaphragm to the image plane, and PZ is a distance from the diaphragm to the thin aspherical plate.

(81) The numerical ranges of the conditional expressions (10) and (11) may be set to those of the following conditional expressions (10a) and (11a).

$$0.1 < PZ/LP < 0.7 \quad (10a)$$

$$0.1 < PZ/LS < 0.7 \quad (11a)$$

(82) The optical systems according to Examples 5 to 8 may satisfy at least one of the following conditional expressions (12) to (14):

$$0.1 < |f1/f| < 5.0 \quad (12)$$

$$0.1 < f2/f < 10.0 \quad (13)$$

$$0.1 < f3/f < 10.0 \quad (14)$$

(83) The conditional expressions (12) to (14) are set to the optical systems according to Examples 5 to 8 so as to satisfactorily correct various aberrations such as the chromatic aberration, the spherical aberration, and the curvature of field and to exhibit high optical performance. If the value is higher than the upper limit or lower than the lower limit in each of the conditional expressions (12) to (14), the correction balance between the curvature of field and the spherical aberration destroys, and the optical performance is deteriorated.

(84) The numerical ranges of the conditional expressions (12) to (14) may be set to those of the following conditional expressions (12a) to (14a).

$$0.1 < |f1/f| < 3.5 \quad (12a)$$

$$0.1 < f2/f < 7.0 \quad (13a)$$

0.1 < f3/f < 4.0 (14a)

(85) Tables 13 and 14 show numerical values corresponding to the conditional expressions according to Examples 5 to 8.

(86) TABLE-US-00013 TABLE 13 f f1 f2 f3 Pf1 f1/f f2/f f3/f f/Pf EX. 5 4.5 -11.68 17.91 14.15 81.34 -2.60 3.98 3.14 0.055 EX. 6 4.5 -10.34 16.23 13.47 72.62 -2.30 3.61 2.99 0.062 EX. 7 3 -57.95 18.257 9.4242 31.166 -1.93 6.09 3.14 0.096 EX. 8 6 -13.09 22.268 12.316 28.474 -2.18 3.71 2.05 0.211

(87) TABLE-US-00014 TABLE 14 LP LS PZ PZ/LP PZ/LS T EX. 5 15.05 19.95 8381 0.59 — 0.20 EX. 6 16.69 18.3 10.39 — 0.57 0.22 EX. 7 7.23 19.73 12.01 — 0.61 0.33 EX. 8 8.54 23.74 15.79 — 0.67 0.45

Example 6

(88) FIG. 11 is a sectional view of an optical system 600 according to this example. The optical system 600 is an infrared optical system having a focal length of 4.5 mm and an F-number of 0.8. The optical system 600 includes, in order from the object side to the image side, a first lens L61 made of a silicon material and having a negative refractive power, a second lens L62 made of a silicon material and having a positive refractive power, a diaphragm S6, a third lens L63 having a positive refractive power and made of a silicon material, and a thin aspherical plate P61 made of a silicon material and having an aspherical portion that is the thinnest between the center and the outermost periphery (not inclusive). Light in the infrared range guided by the optical system 600 passes through a cover glass CG6 and forms an image on an infrared sensor IM6. Table 15 shows numerical data of the optical system 600. The unit of radius of curvature and spacing is mm.

(89) TABLE-US-00015 TABLE 15 RADIUS OF GLASS CURVATURE SPACING MATERIAL OBJECT — INFINITY PLANE L61 SPHERICAL 9.46 0.45 SILICON SURFACE SPHERICAL 6.63 14.65 SURFACE L62 SPHERICAL 49.33 1.50 SILICON SURFACE SPHERICAL -186.71 0.10 SURFACE S6 FLAT FLAT 7.51 SURFACE SURFACE L63 SPHERICAL 14.94 1.70 SILICON SURFACE SPHERICAL 25.41 1.18 SURFACE P61 ASPHERICAL 175.36 0.22 SILICON SURFACE 61 SPHERICAL FLAT 2.00 SURFACE SURFACE CG6 FLAT FLAT 1.00 GER- SURFACE SURFACE MANIUM FLAT FLAT 4.68 SURFACE SURFACE IM6 — —

(90) Table 16 shows aspherical shape data. The aspherical shape is represented by the expression (1).

(91) TABLE-US-00016 TABLE 16 ASPHERICAL SURFACE 61 PARAXIAL RADIUS OF CURVATURE R 175.359 CONICAL COEFFICIENT k 0.00 4TH-ORDER COEFFICIENT A -6.07E-05 6TH-ORDER COEFFICIENT B 3.43E-07 8TH-ORDER COEFFICIENT C 0.00E+00 10TH-ORDER COEFFICIENT D 0.00E+00 12TH-ORDER COEFFICIENT E 0.00E+00 14TH-ORDER COEFFICIENT F 0.00E+00

(92) FIG. 12 is an MTF diagram of an optical system 600. In this example, an infrared sensor IM6 having a pixel pitch of 17 μm is used as an example. In this case, the Nyquist frequency is about 30 lp/mm. The MTF value at the Nyquist frequency of this example represented by reference numeral 61 in FIG. 12 is 50%. Since the MTF value is 30% or higher at the Nyquist frequency of the infrared sensor IM6, the optical system 600 exhibits excellent optical performance.

Example 7

(93) FIG. 13 is a sectional view of an optical system 700 according to this example. The optical system 700 is an infrared optical system having a focal length of 3 mm and an F-number of 0.8. The optical system 700 includes, in order from the object side to the image side, a first lens L71 having a negative refractive power and made of a silicon material, a diaphragm S7, a second lens L72 having a positive refractive power and made of a silicon material, a third lens L73 having a positive refractive power and made of a silicon material, and a thin aspherical plate P71 made of a silicon material and having an aspherical portion that is the thinnest between the center and the outermost periphery (not inclusive). Light in the infrared range guided by the optical system 700 passes through a cover glass CG7 and forms an image on an infrared sensor IM7. Table 17 shows

numerical data of the optical system **700**. The unit of radius of curvature and spacing is mm.

(94) TABLE-US-00017 TABLE 17 RADIUS OF GLASS CURVATURE SPACING MATERIAL OBJECT — INFINITY PLANE L71 SPHERICAL 9.72 0.45 SILICON SURFACE SPHERICAL 7.43 10.05 SURFACE S7 SPHERICAL FLAT 0.10 SURFACE SURFACE L72 SPHERICAL 19.87 1.00 SILICON SURFACE FLAT 28.08 8.70 SURFACE L73 SPHERICAL 15.16 2.00 SILICON SURFACE SPHERICAL 29.26 2.62 SURFACE P71 ASPHERICAL 83.69 0.22 SILICON SURFACE 71 SPHERICAL FLAT 2.00 SURFACE SURFACE CG7 FLAT FLAT 1.00 GERMANIUM SURFACE SURFACE FLAT FLAT 4.50 SURFACE SURFACE IM7 — —

(95) Table 18 shows aspherical shape data. The aspherical shape is represented by the expression (1).

(96) TABLE-US-00018 TABLE 18 ASPHERICAL SURFACE 71 PARAXIAL RADIUS OF CURVATURE R 75.259 CONICAL COEFFICIENT k 0.00 4TH-ORDER COEFFICIENT A $-1.35\text{E}-04$ 6TH-ORDER COEFFICIENT B $6.62\text{E}-07$ 8TH-ORDER COEFFICIENT C 0.00E+00 10TH-ORDER COEFFICIENT D 0.00E+00 12TH-ORDER COEFFICIENT E 0.00E+00 14TH-ORDER COEFFICIENT F 0.00E+00

(97) FIG. **14** is an MTF diagram of the optical system **700**. This example uses the infrared sensor IM7 having a pixel pitch of 17 μm as an implementation. In this case, the Nyquist frequency is about 30 lp/mm. The MTF value at the Nyquist frequency in this example represented by reference numeral **71** in FIG. **14** is 49%. Since the MTF value is 30% or higher at the Nyquist frequency of the infrared sensor IM7, the optical system **700** exhibits excellent optical performance.

Example 8

(98) FIG. **15** is a sectional view of an optical system **800** according to this example. The optical system **800** is an infrared optical system having a focal length of 6 mm and an F-number of 0.8. The optical system **800** includes, in order from the object side to the image side, a first lens L81 having a negative refractive power and made of a silicon material, a diaphragm S8, a second lens L82 having a positive refractive power and made of a silicon material, a third lens L83 having a positive refractive power and made of a silicon material, and a thin aspherical plate P81 made of a silicon material and having an aspherical portion that is the thinnest between the center and the outermost periphery (not inclusive). Light in the infrared range guided by the optical system **800** passes through a cover glass CG8 and forms an image on an infrared sensor IM8. Table 19 shows numerical data of the optical system **800**. The unit of radius of curvature and spacing is mm.

(99) TABLE-US-00019 TABLE 19 RADIUS OF GLASS CURVATURE SPACING MATERIAL OBJECT — INFINITY PLANE L81 SPHERICAL 10.79 0.45 SILICON SURFACE SPHERICAL 7.81 8.09 SURFACE S8 SPHERICAL FLAT 0.10 SURFACE SURFACE L82 SPHERICAL 24.50 1.00 SILICON SURFACE FLAT SURFACE 43.71 9.98 L83 SPHERICAL 16.13 2.00 SILICON SURFACE SPHERICAL 32.18 2.71 SURFACE P81 ASPHERICAL 68.76 0.45 SILICON SURFACE 81 SPHERICAL FLAT 2.00 SURFACE SURFACE CG8 FLAT SURFACE FLAT 1.00 GER- SURFACE MANIUM PLAT SURFACE FLAT 4.50 SURFACE IM8 — —

(100) Table 20 shows aspherical shape data. The aspherical shape is represented by the expression (1).

(101) TABLE-US-00020 TABLE 20 ASPHERICAL SURFACE 81 PARAXIAL RADIUS OF CURVATURE R 68.759 CONICAL COEFFICIENT k 0.00 4TH-ORDER COEFFICIENT A $-1.02\text{E}-04$ 6TH-ORDER COEFFICIENT B $3.50\text{E}-07$ 8TH-ORDER COEFFICIENT C 0.00E+00 10TH-ORDER COEFFICIENT D 0.00E+00 12TH-ORDER COEFFICIENT E 0.00E+00 14TH-ORDER COEFFICIENT F 0.00E+00

(102) FIG. **16** is an MTF diagram of an optical system **800**. This example uses the infrared sensor IM8 having a pixel pitch of 17 μm as an implementation. In this case, the Nyquist frequency is about 30 lp/mm. The MTF value at the Nyquist frequency in this example represented by reference numeral **81** in FIG. **16** is 54%. Since the MTF value is 30% or higher at the Nyquist frequency of the infrared sensor IM8, the optical system **800** exhibits excellent optical performance.

Example 9

(103) FIG. 17 is a sectional view of an optical system 900 according to this example. The optical system 900 is an infrared optical system having a focal length of 4.5 mm and an F-number of 0.8. The optical system 900 includes, in order from the object side to the image side, a first lens L91 having a negative refractive power and made of a germanium material, a second lens L92 having a positive refractive power and made of a silicon material, a diaphragm S9, a third lens L93 having a positive refractive power and made of a silicon material, and an optical element (aspherical lens) ASP91 having an aspherical surface and made of a chalcogenide material. Chalcogenide contains a compound using an oxygen group element of Group 16 in the periodic table, and the refractive index and Abbe number change depending on the type of the compound. In this example, IRG206 of NHG (Hubei New Huagang Information Materials Co., Ltd.) is used as chalcogenide, but the compound is not limited to this example as long as it is a compound using an oxygen group element of Group 16 in the periodic table. Light in the infrared range guided by the optical system 800 passes through a cover glass CG9 and forms an image on an infrared sensor IM9. Table 21 shows numerical data of the optical system 900. The unit of radius of curvature and spacing is mm.

(104) TABLE-US-00021 TABLE 21 RADIUS OF GLASS CURVATURE SPACING MATERIAL OBJECT PLANE — INFINITY L91 11.4429 0.45 GERMANIUM 8.208534 20.08 L92 33.64908 1.50 SILICON 103.3151 0.96 S9 FLAT SURFACE 10.01 L93 12.39436 1.70 SILICON 17.8448 2.07 ASP91(ASPHERICAL 358.0504 1.00 CALCOGENIDE SURFACE 91) FLAT SURFACE 2.00 CG9 FLAT SURFACE 1.00 GERMANIUM FLAT SURFACE 4.56 IM9 —

(105) Table 22 shows aspherical shape data. The aspherical shape is represented by the expression (1).

(106) TABLE-US-00022 TABLE 22 ASPHERICAL SURFACE 91 PARAXIAL RADIUS OF CURVATURE R 358.050 CONICAL COEFFICIENT k 0.00 4TH-ORDER COEFFICIENT A $-1.05\text{E}-04$ 6TH-ORDER COEFFICIENT B $2.95\text{E}-06$ 8TH-ORDER COEFFICIENT C $-1.13\text{E}-07$ 10TH-ORDER COEFFICIENT D $3.21\text{E}-09$ 12TH-ORDER COEFFICIENT B $-4.79\text{E}-11$ 14TH-ORDER COEFFICIENT F $2.81\text{E}-13$

(107) FIG. 18 is an MTF diagram of the optical system 900. This example uses the infrared sensor IM9 having a pixel pitch of 17 pin as an implementation. In this case, the Nyquist frequency is about 30 lp/mm. The MTF value at the Nyquist frequency in this example represented by reference numeral 91 in FIG. 18 is 57%. Since the MTF value is 30% or higher at the Nyquist frequency of the infrared sensor IM9, the optical system 900 exhibits excellent optical performance.

(108) Silicon materials and germanium materials each have a high refractive index and a low dispersion in the infrared range and can exhibit high optical performance using an aspherical surface used for part of the lens made of silicon material or germanium material. However, in order to process an optical element having an aspherical surface of a silicon material or a germanium material, a highly difficult process such as grinding or polishing is required.

(109) On the other hand, chalcogenide, zinc selenium (ZnSe), zinc sulfide (ZnS), resin (high density polyethylene) and the like are materials that can be molded by heat or the like, but exhibits a large chromatic aberration amount due to a large dispersion. If a diffraction structure is provided on the lens surface in order to correct the chromatic aberration, unnecessary light may be incident on the infrared sensor due to scattering in the diffraction structure, which may deteriorate the optical performance.

(110) A lens usually has a positive Abbe number, and thus at least one lens may have a negative focal distance in order to reduce the chromatic aberration. Using materials with a large dispersion such as chalcogenide, zinc selenium, zinc sulfide, and resin (high density polyethylene) can suppress the chromatic aberration by increasing the focal distance of the lens (or by reducing the refractive power).

(111) One method of correcting the curvature of field with high accuracy is a method of placing an optical element having an aspherical surface at a position where light rays of respective angles of

view are separated. In an attempt to mainly correct the curvature of field, it is effective to dispose an aspherical lens on the image side. When such a moldable material as chalcogenide, zinc selenide, zinc sulfide, and resin (high density polyethylene) is used for the aspherical lens, the chromatic aberration becomes a problem. In order to solve this problem, an aspherical lens having a long focal length is used in this example.

(112) Thus, this example can improve optical performance by disposing the optical element having an aspherical surface with a small refractive power at a position different from that of the diaphragm. The optical element having an aspherical surface may satisfy the conditional expression of $20 \leq \nu_{10} \leq 2000$. In order to improve the ease of manufacturing (molding and processing) an optical element having an aspherical surface, the optical element may be made of a material having a large dispersion. More specifically, the optical element having an aspherical surface may satisfy the conditional expression of $20 \leq \nu_{10} \leq 800$.

(113) Compared to aspherical lenses, aspherical lenses using materials with large dispersion need to suppress the chromatic aberration while giving only the action of the aspherical part as an effect. Therefore, the following expression (15) may be satisfied:

$$0.0 < |f/Pf_2| < 0.3 \quad (15)$$

where Pf_2 is a focal length of the aspherical lens.

(114) The numerical range of the conditional expression (15) may be set to that of the following conditional expression (15a).

$$0.0 < |f/Pf_2| < 0.1 \quad (15a)$$

(115) The numerical range of the conditional expression (15) may be set to that of the following conditional expression (15b).

$$0.00 < |f/Pf_2| < 0.07 \quad (15b)$$

(116) The aspherical lens using a material having a large dispersion may be disposed on the image side of the diaphragm from the viewpoint of the curvature of field correction. The following expression (16) may be satisfied:

$$0.3 < PZ/LS < 1.0 \quad (16)$$

where PZ is a distance from the diaphragm to the aspherical lens.

(117) The numerical range of the conditional expression (16) may be set to that of the following conditional expression (16a).

$$0.4 < PZ/LS < 0.9 \quad (16a)$$

(118) The optical systems according to Examples 9 to 12 may satisfy at least one of the following conditional expressions (17) to (19).

$$0.01 < |f_1/Pf_2| < 0.80 \quad (17)$$

$$0.01 < |f_2/Pf_2| < 0.80 \quad (18)$$

$$0.01 < |f_3/Pf_2| < 0.80 \quad (19)$$

(119) The conditional expressions (17) to (19) are set to the optical systems according to Examples 9 to 12 so as to satisfactorily correct various aberrations such as the chromatic aberration, the spherical aberration, and the curvature of field and to exhibit high optical performance. If the value is higher than the upper limit or lower than the lower limit in each of the conditional expressions (17) to (19), the correction balance between the curvature of field and the spherical aberration destroys, and the optical performance is deteriorated.

(120) The numerical ranges of the conditional expressions (17) to (19) may be set to those of the following conditional expressions (17a) to (19a).

$$0.01 < |f_1/Pf_2| < 0.50 \quad (17a)$$

$$0.01 < |f_2/Pf_2| < 0.50 \quad (18a)$$

$$0.01 < |f_3/Pf_2| < 0.50 \quad (19a)$$

(121) Tables 23 and 24 show numerical values corresponding to the conditional expressions according to Examples 9 to 12.

(122) TABLE-US-00023 TABLE 23 f f1 f2 f3 Pf f1/f f2/f f3/f f/Pf EX. 9 4.5 -9.95 18.15 13.60

80.16 -2.21 4.03 3.02 0.050 EX. 10 4.5 -10.24 17.92 13.54 130.83 -2.28 3.98 3.01 0.034 EX. 11
4.5 -9.834 17.78 13.538 140.82 -2.19 3.95 3.01 0.032 EX. 12 4.5 -10.66 17.884 13.547 203.53
-2.37 3.97 3.01 0.022

(123) TABLE-US-00024 TABLE 24 LS PZ PZ/LS EX. 9 22.34 13.78 0.62 EX. 10 21.05 12.54
0.60 EX. 11 21.45 12.74 0.59 EX. 12 20.4 12.24 0.60

Example 10

(124) FIG. **19** is a sectional view of an optical system **1000** according to this example. The optical system **1000** is an infrared optical system having a focal length of 4.5 mm and an F-number of 0.8. The optical system **1000** includes, in order from the object side to the image side, a first lens **L101** having a negative refractive power and made of a silicon material, a second lens **L102** having a positive refractive power and made of a silicon material, a diaphragm **S10**, a third lens **L103** having a positive refractive power and made of a silicon material, and an optical element (aspherical lens) **ASP101** having an aspherical surface and made of a zinc sulfide material. Light in the infrared range guided by the optical system **1000** passes through a cover glass **CG10** and forms an image on an infrared sensor **IM10**. Table 25 shows numerical data of the optical system **1000**. The unit of radius of curvature and spacing is mm.

(125) TABLE-US-00025 TABLE 25 RADIUS OF GLASS CURVATURE SPACING MATERIAL
OBJECT PLANE — INFINITY L101 10.8801 0.45 SILICON 7.33552 16.85 L102 52.42293 1.50
SILICON -242.625 0.10 S10 FLAT SURFACE 9.32 L103 14.17181 1.70 SILICON 22.89453 1.53
ASP101(ASPHERICAL 151.7479 1.00 ZINC SURFACE 101) SULFIDE 1.73E+04 2.00 CG10
FLAT SURFACE 1.00 GER- MANIUM FLAT SURFACE 4.56 IM10 —

(126) Table 26 shows aspherical shape data. The aspherical shape is represented by the expression (1).

(127) TABLE-US-00026 TABLE 26 ASPHERICAL SURFACE 101 PARAXIAL RADIUS OF
CURVATURE R 151.748 CONICAL COEFFICIENT k 0.00 4TH-ORDER COEFFICIENT A
-1.34E-04 6TH-ORDER COEFFICIENT B 4.16E-06 8TH-ORDER COEFFICIENT C
-1.68E-07 10TH-ORDER COEFFICIENT D 4.26E-09 12TH-ORDER COEFFICIENT E
-5.50E-11 14TH-ORDER COEFFICIENT F 2.81E-13

(128) FIG. **20** is an MTF diagram of the optical system **1000**. This example uses the infrared sensor **IM10** having a pixel pitch of 17 μm as an implementation. In this case, the Nyquist frequency is about 30 lp/mm. The MTF value at the Nyquist frequency in this example represented by reference numeral **101** in FIG. **20** is 52%. Since the MTF value is 30% or higher at the Nyquist frequency of the infrared sensor **IM10**, the optical system **1000** exhibits excellent optical performance.

Example 11

(129) FIG. **21** is a sectional view of an optical system **1100** according to this example. The optical system **1100** is an infrared optical system having a focal length of 4.5 mm and an F-number of 0.8. The optical system **1100** includes, in order from the object side to the image side, a first lens **L111** having a negative refractive power and made of a silicon material, a second lens **L112** having a positive refractive power and made of a silicon material, a diaphragm **S11**, a third lens **L113** having a positive refractive power and made of a silicon material, and an optical element (aspherical lens) **ASP111** having an aspherical surface and made of a zinc selenium material. Light in the infrared range guided by the optical system **1100** passes through a cover glass **CG11** and forms an image on an infrared sensor **IM11**. Table 27 shows numerical data of the optical system **1100**. The unit of radius of curvature and spacing is mm.

(130) TABLE-US-00027 TABLE 27 RADIUS OF GLASS CURVATURE SPACING MATERIAL
OBJECT PLANE — INFINITY L111 11.24434 0.45 SILICON 7.41501 16.50 L112 42.69496 1.50
SILICON 7444.902 0.10 S11 FLAT SURFACE 9.50 L113 14.16753 1.70 SILICON 22.88135 1.54
ASP111(ASPHERICAL 154.7086 1.00 ZINC SURFACE 111) SELENIDE 7.29E+02 2.00 CG11
FLAT SURFACE 1.00 GER- MANIUM FLAT SURFACE 4.70 IM11 —

(131) Table 28 shows aspherical shape data. The aspherical shape is represented by the expression

(1).

(132) TABLE-US-00028 TABLE 28 ASPHERICAL SURFACE 111 PARAXIAL RADIUS OF CURVATURE R 154.709 CONICAL COEFFICIENT k 0.00 4TH-ORDER COEFFICIENT A $-1.23\text{E}-04$ 6TH-ORDER COEFFICIENT B $3.58\text{E}-06$ 8TH-ORDER COEFFICIENT C $-1.48\text{E}-07$ 10TH-ORDER COEFFICIENT D $3.92\text{E}-09$ 12TH-ORDER COEFFICIENT E $-5.29\text{E}-11$ 14TH-ORDER COEFFICIENT F $2.81\text{E}-13$

(133) FIG. 22 is an MTF diagram of the optical system **1100**. This example uses the infrared sensor IM10 having a pixel pitch of 17 μm as an implementation. In this case, the Nyquist frequency is about 30 lp/mm. The MTF value at the Nyquist frequency in this example represented by reference numeral **111** in FIG. 22 is 57%. Since the MTF value is 30% or higher at the Nyquist frequency of the infrared sensor IM10, the optical system **1100** exhibits excellent optical performance.

Example 12

(134) FIG. 23 is a sectional view of an optical system **1200** according to this example. The optical system **1200** is an infrared optical system having a focal length of 4.5 mm and an F-number of 0.8. The optical system **1200** includes, in order from the object side to the image side, a first lens L121 having a negative refractive power and made of a silicon material, a second lens L122 having a positive refractive power and made of a silicon material, a diaphragm S12, a third lens L123 having a positive refractive power and made of a silicon material, and an optical element (aspherical lens) ASP121 having an aspherical surface made of a resin material. A resin material amount includes high-density polyethylene (HDPE), which is polyethylene having a density of 0.942 [kg/m^3] or higher. Light in the infrared range guided by the optical system **1200** passes through a cover glass CG12 and forms an image on an infrared sensor IM12. Table 29 shows numerical data of the optical system **1200**. The unit of radius of curvature and spacing is mm.

(135) TABLE-US-00029 TABLE 29 RADIUS OF GLASS CURVATURE SPACING MATERIAL OBJECT PLANE — INFINITY L121 10.34174 0.45 SILICON 7.150503 17.55 L122 83.9322 1.50 SILICON -87.8344 0.10 S12 FLAT SURFACE 8.88 L123 14.08102 1.70 SILICON 22.61232 1.66 ASP121(ASPHERICAL 120.1702 1.00 RESIN SURFACE 121) $-7.64\text{E}+02$ 2.00 CG12 FLAT SURFACE 1.00 GERMANIUM FLAT SURFACE 4.16 IM12 —

(136) Table 30 shows aspherical shape data. The aspherical shape is represented by the expression (1).

(137) TABLE-US-00030 TABLE 30 ASPHERICAL SURFACE 121 PARAXIAL RADIUS OF CURVATURE R 120.170 CONICAL COEFFICIENT k 0.00 4TH-ORDER COEFFICIENT A $-2.77\text{E}-04$ 6TH-ORDER COEFFICIENT B $1.14\text{E}-05$ 8TH-ORDER COEFFICIENT C $-4.22\text{E}-07$ 10TH-ORDER COEFFICIENT D $8.76\text{E}-09$ 12TH-ORDER COEFFICIENT B $-8.50\text{E}-11$ 14TH-ORDER COEFFICIENT F $2.81\text{E}-13$

(138) FIG. 24 is an MTF diagram of the optical system **1200**. This example uses the infrared sensor IM7 having a pixel pitch of 17 μm as an implementation. In this case, the Nyquist frequency is about 30 lp/mm. The MTF value at the Nyquist frequency in this example represented by reference numeral **121** in FIG. 24 is 43%. Since the MTF value is 30% or higher at the Nyquist frequency of the infrared sensor IM12, the optical system **1200** exhibits excellent optical performance.

Example 13

(139) In this example, an infrared camcorder (video camera) is an illustrative image pickup apparatus using the optical system according to each example. FIG. is a schematic view of principal part of the camcorder according to this example. The camcorder has an imaging optical system **11** including a camera body **13** and an optical system according to any one of Examples 1 to 12. The camera body **13** includes an image sensor (infrared sensor) **12** such as a microbolometer that receives (photoelectrically converts) an object image formed by the imaging optical system **11**. The infrared sensor can use, for example, one formed by using vanadium oxide or amorphous silicon. An acquired image can be confirmed on a display device **14**. The display device **14** may be incorporated into the camera body **13** or may be separated from but communicable with the camera

body 13 via wireless communications. The optical system according to each example is applicable to an image pickup apparatus such as an in-vehicle camera or a surveillance camera.

(140) Each example can provide an optical system having high optical performance in the infrared range and easy to manufacture, and an image pickup apparatus having the same.

(141) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(142) This application claims the benefit of Japanese Patent Application No. 2020-108208, filed on Jun. 23, 2020, which is hereby incorporated by reference herein in its entirety.

Claims

1. An optical system configured to image an object with light with a wavelength of 8 μm or longer, the optical system comprising: a first lens having a negative refractive power; a second lens having a positive refractive power and closer to an image plane than the first lens; a third lens having a positive refractive power and closer to the image plane than the second lens; a diaphragm; and an optical element having an aspherical surface and disposed at a position different from that of the diaphragm, wherein in a section including an optical axis, a thickness of the optical element is minimum at a position other than an on-axis and an outermost off-axis, and wherein the following conditional expression is satisfied:

$0.0 < |f/Pf1| < 0.3$ where f is a focal length of the optical system, and $Pf1$ is a focal length of the optical element; wherein the following conditional expression is satisfied:

$0.1 < |f1/f| < 5.0$ where $f1$ is a focal length of the first lens.

2. The optical system according to claim 1, wherein the aspherical surface has a shape represented by a function of a sixth order or longer.

3. The optical system according to claim 1, wherein the optical element is disposed on an object side of the diaphragm, and wherein the following conditional expression is satisfied:

$PZ/LP < 0.8$ where PZ is a distance from the diaphragm to the optical element, and LP is a distance from the diaphragm to the first lens.

4. The optical system according to claim 1, wherein the optical element is closer to the image plane than the diaphragm, and wherein the following conditional expression is satisfied:

$PZ/LS < 0.8$ where PZ is a distance from the diaphragm to the optical element, and LS is a distance from the diaphragm to the image plane.

5. The optical system according to claim 1, wherein the following conditional expression is satisfied:

$0.1 < f2/f < 10.0$ where $f2$ is a focal length of the second lens.

6. The optical system according to claim 1, wherein the following conditional expression is satisfied:

$0.05 \leq T \leq 1.00$ where T [mm] is a thickness of the optical element on an optical axis.

7. The optical system according to claim 1, wherein the optical element is closer to the image plane than the diaphragm, and wherein the following conditional expression is satisfied:

$0.3 < PZ/LS < 1.0$ where PZ is a distance from the diaphragm to the optical element, and LS is a distance from the diaphragm to the image plane.

8. The optical system according to claim 1, wherein the following conditional expression is satisfied:

$0.01 < |f1/Pf1| < 0.80$ where $f1$ is a focal length of the first lens.

9. The optical system according to claim 1, wherein the following conditional expression is satisfied:

$0.01 < f2/Pf1 < 0.80$ where $f2$ is a focal length of the second lens.

10. The optical system according to claim 1, wherein the following conditional expression is satisfied:

$0.1 < f_3/Pf_1 < 0.80$ where f_3 is a focal length of the third lens.

11. The optical system according to claim 1, wherein the optical element is made of a silicon material.

12. The optical system according to claim 1, each of the first lens, the second lens, and third lens is made of a silicon material or a germanium material.

13. An image pickup apparatus comprising: the optical system according to claim 1; and an image sensor configured to receive light from the optical system.

14. An optical system configured to image an object with light with a wavelength of 8 μm or longer, the optical system comprising: a first lens having a negative refractive power; a second lens having a positive refractive power and closer to an image plane than the first lens; a third lens having a positive refractive power and closer to the image plane than the second lens; a diaphragm; and an optical element having an aspherical surface and disposed at a position different from that of the diaphragm, wherein in a section including an optical axis, a thickness of the optical element is minimum at a position other than an on-axis and an outermost off-axis, and wherein the following conditional expression is satisfied:

$0.0 < |f/Pf_1| < 0.3$ where f is a focal length of the optical system, and Pf_1 is a focal length of the optical element, and wherein the following conditional expression is satisfied:

$0.1 < f_3/f < 10.0$ where f_3 is a focal length of the third lens.

15. An image pickup apparatus comprising: the optical system according to claim 14; and an image sensor configured to receive light from the optical system.
